Code No: 136FF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
INDUSTRIAL MANAGEMENT
(Common to ME, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the key components of Taylor's Scientific Management Theory?	[2]
b)	List any six principles of management.	[3]
c)	Explain the differences between line organization, line and staff organization.	[2]
d)	Define departmentation and decentralization.	[3]
e)	Define mass production.	[2]
f)	Describe the product design process and its key stages.	[3]
g)	Define work measurement.	[2]
h)	What are the key steps involved in conducting a work study?	[3]
i)	Define job evaluation.	[2]
j)	What are the different methods of job evaluation?	[3]

PART – B

(50 Marks)

2. Explain the difference between McGregor's Theory X and Theory Y in terms of assumptions about employee behavior. [10]

OR

3. Evaluate the applicability of Herzberg's Two-Factor Theory of Motivation in today's dynamic work environments, considering its strengths and limitations. [10]

4. How does a matrix organization differ from a committee organization in terms of structure and decision-making processes? [10]

OR

5. How would you apply the concept of a virtual organization to enhance collaboration and flexibility within a geographically dispersed team? [10]

6. Explain the differences between job production, batch production, and mass production systems. [10]

OR

7. Assess the challenges and benefits of implementing a lean production system in a traditional mass production facility, considering factors such as cultural change and resource allocation. [10]

8.a) Define work study and its primary objectives.
b) How does work measurement differ from method study? [5+5]

OR

9.a) Evaluate the effectiveness of work sampling as a method for estimating workloads and identifying process bottlenecks within a healthcare setting.
b) Differentiate between variables and attributes control charts. [5+5]

10. How does the classification method in job evaluation categorize jobs based on predetermined factors or criteria? [10]

OR

11.a) Define PERT and CPM and explain their significance in project management.
b) Consider a software development project with the following activities:

Activity	Optimistic Time (days)	Most Likely Time (days)	Pessimistic Time (days)
A	4	6	8
B	5	7	9
C	3	5	7
D	6	8	10
E	2	4	6
F	4	6	8

Assuming activities are independent, calculate:
i) Expected time for each activity
ii) Variance for each activity
iii) Expected project duration
iv) Probability that the project will be completed within 24 days. [5+5]

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